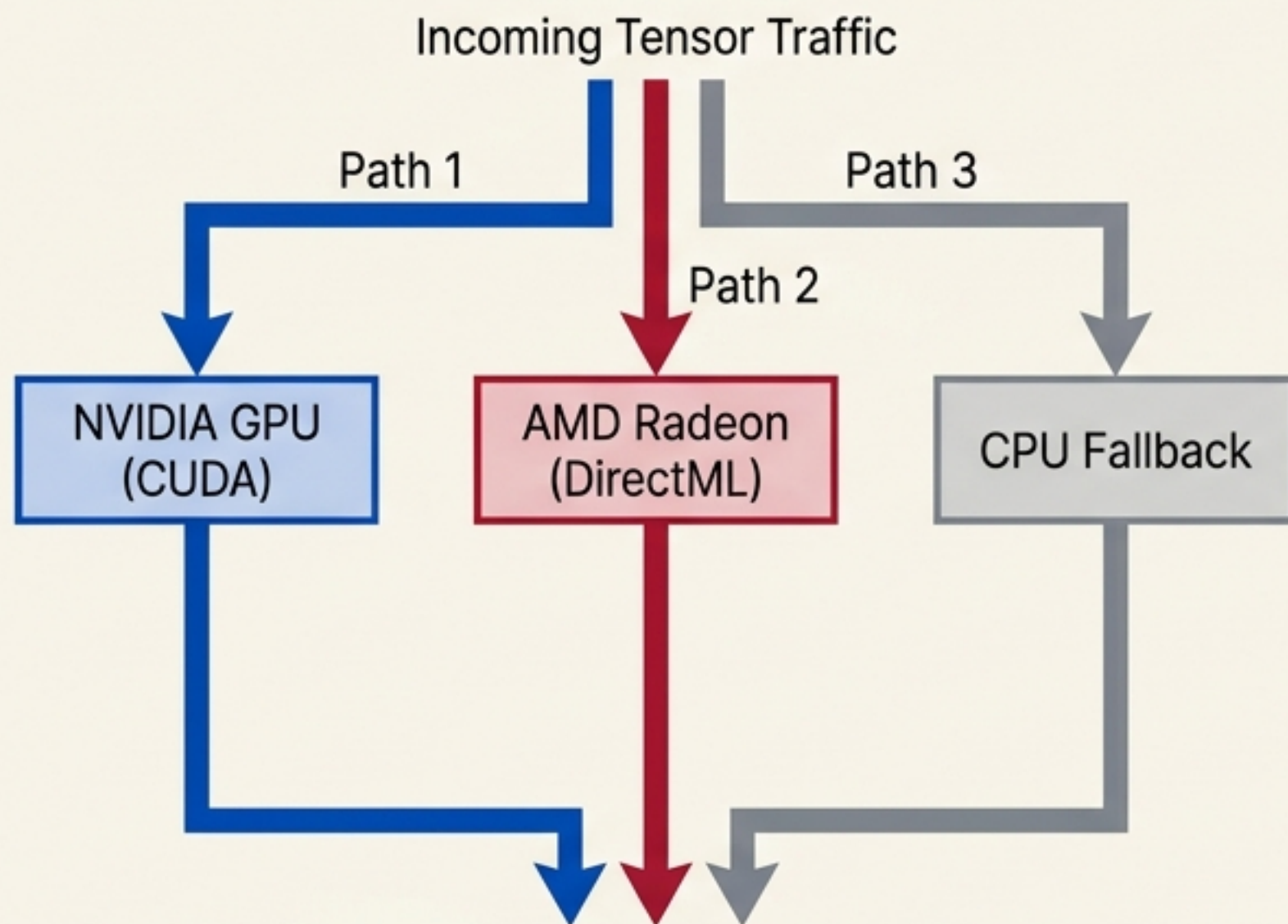


Engineering a Crash-Proof Engine for Hardware Execution

The Hardware Auto-Detection Engine

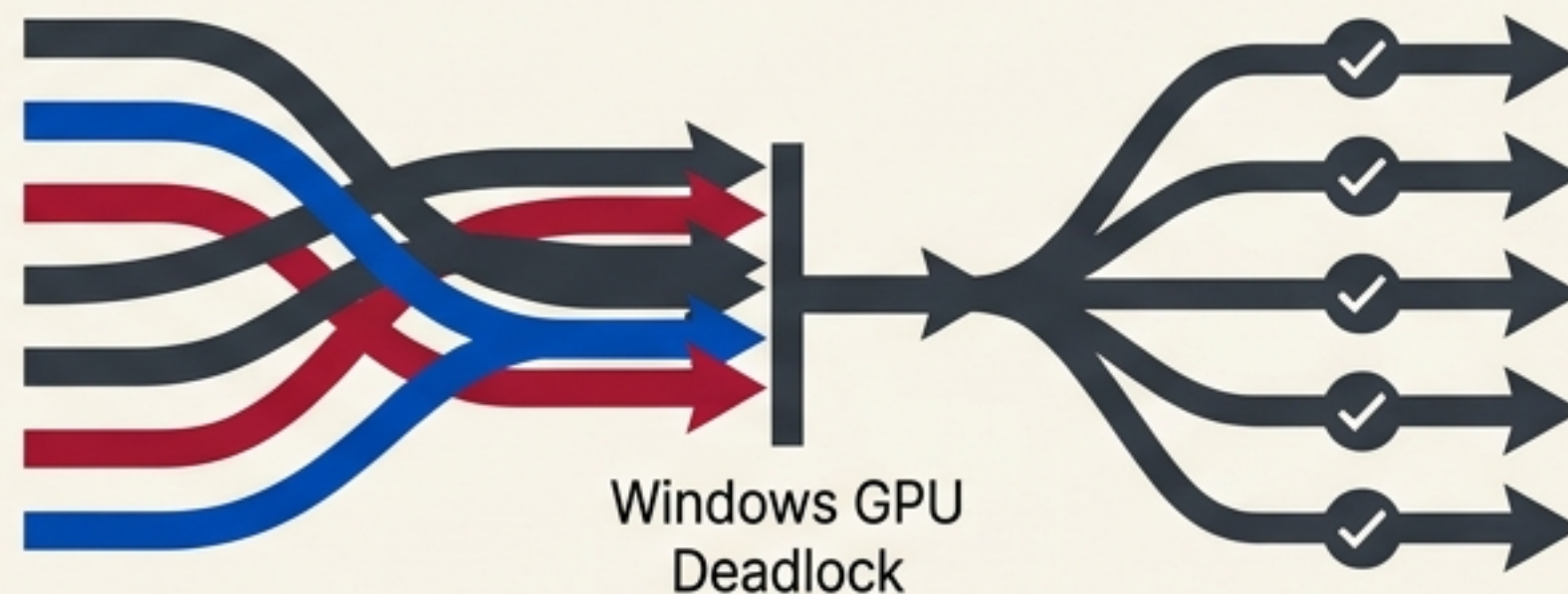


Dynamic Auto-Detection: Real-time tensor routing to NVIDIA CUDA or AMD DirectML ensures 100% self-healing, cross-device execution.

The Concurrency Fix

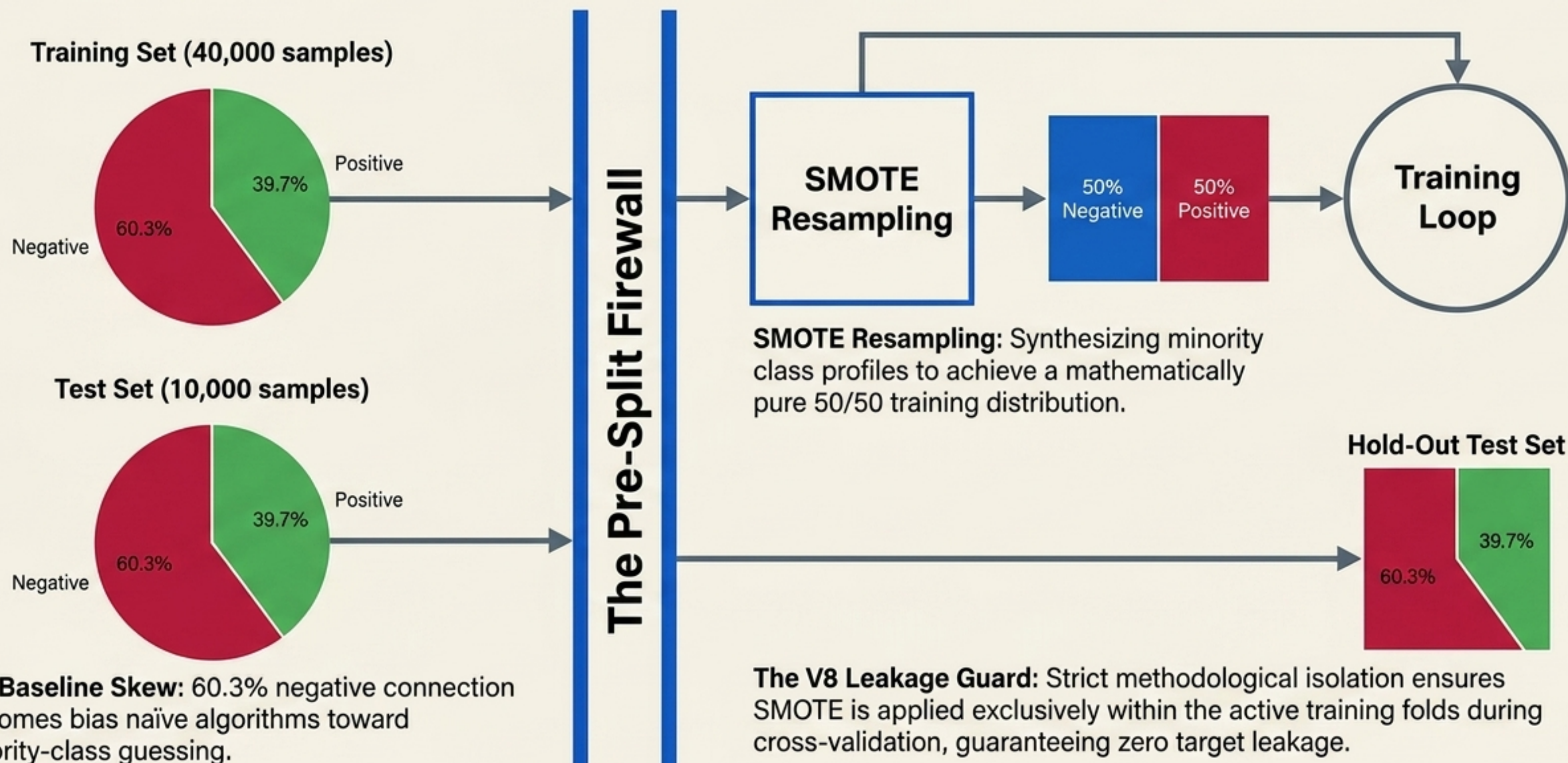
Cross-Validation Loop

WARNING: `n_jobs=-1` → 100% Utilization GPU Crash
FIX APPLIED: `n_jobs=1` (Sequential Outer Isolation)

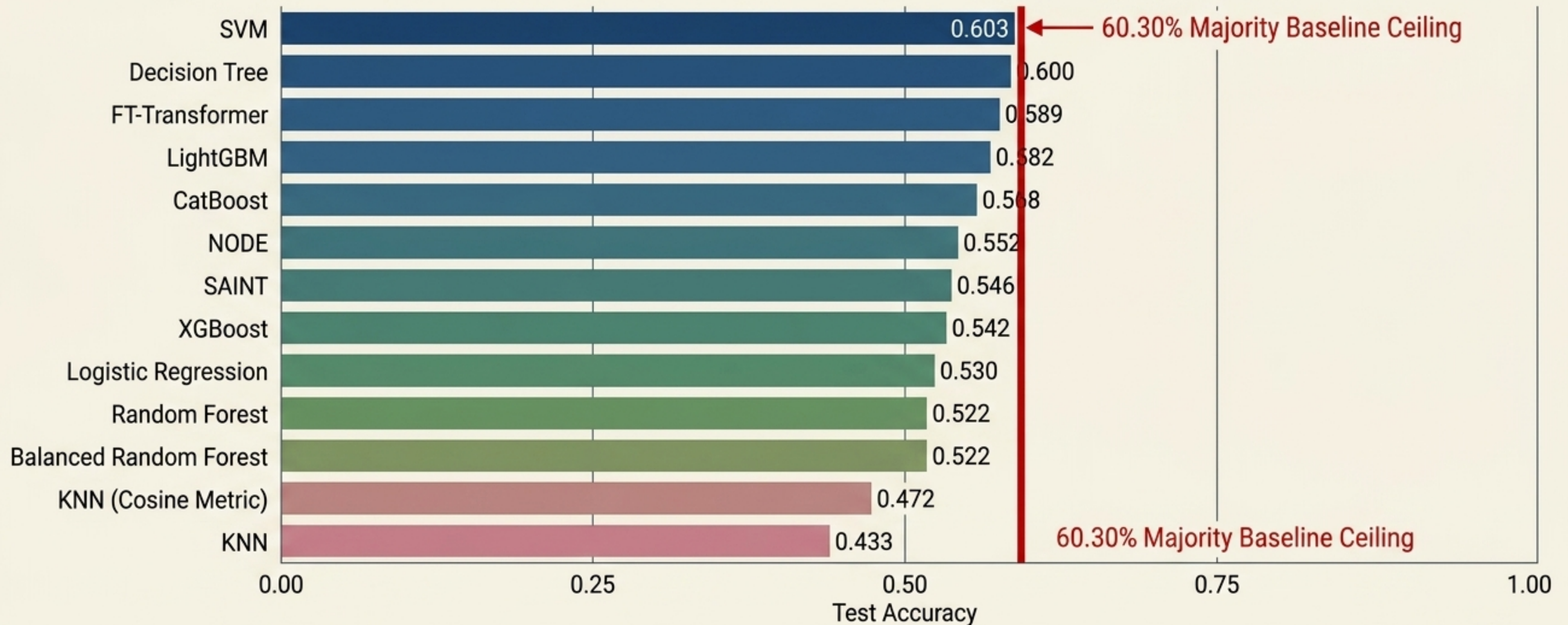


- **The Deadlock Dilemma:** Concurrent GPU context initializations (under `n_jobs=-1`) cause severe driver lock-ups during cross-validation.
- **Sequential Outer Isolation:** Forcing `n_jobs=1` on outer CV loops guarantees clean, sequential GPU access while unlocking **estimators** to fully parallelize internally.

Protecting the Ground Truth of Matchmaking Data



Pushing 14 Classifiers to the Mathematical Limit



AutoML Benchmarking

- PyCaret and FLAML deployed to establish robust, unbiased algorithmic ceilings.

The Traditional Arsenal

12 additional architectures evaluated, covering linear boundaries, boosting ensembles, and multi-threaded SVM bagging.

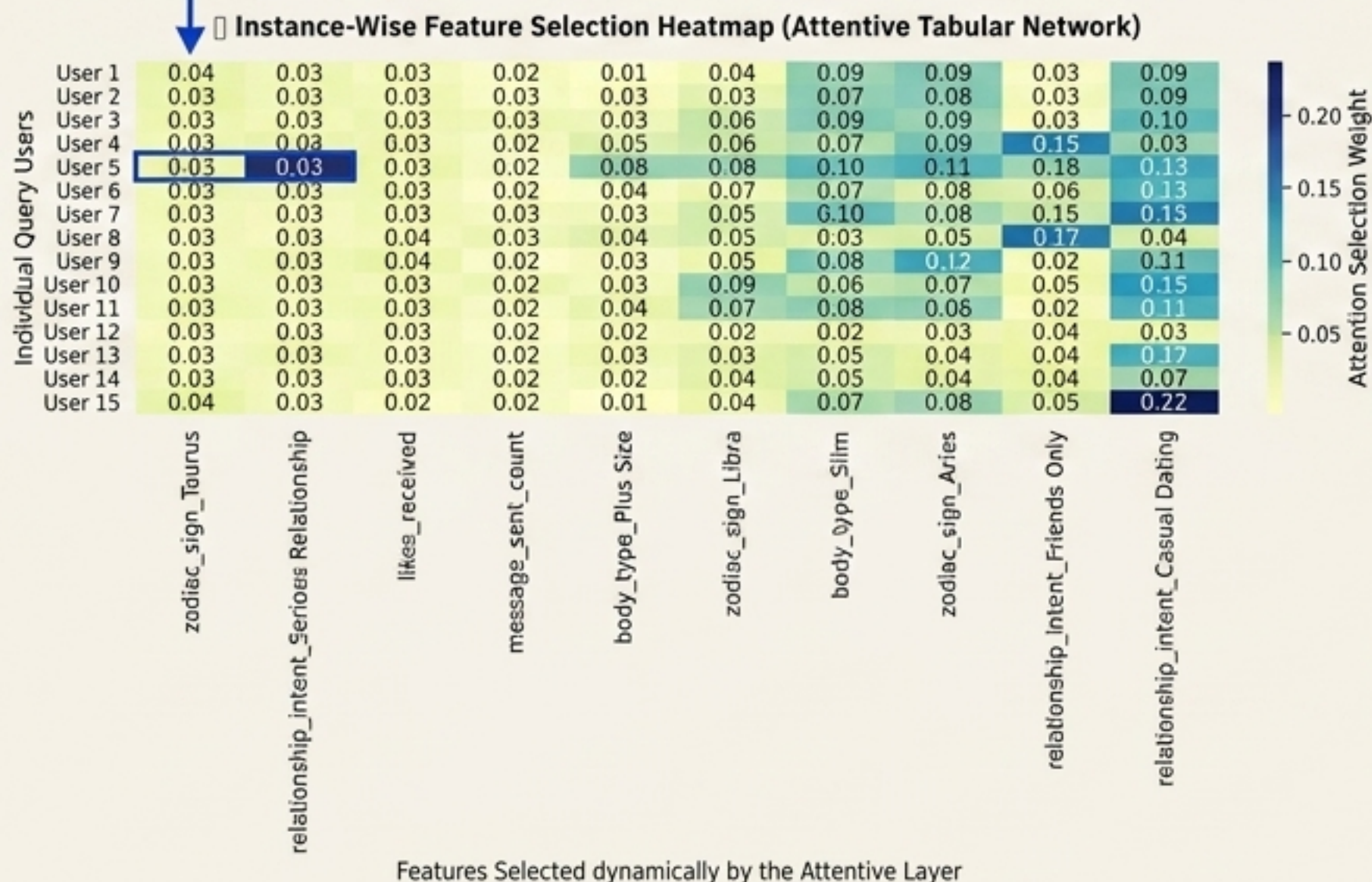
The Convergence Phenomenon

Despite extreme mathematical variances in approach, performance hits a rigid ceiling at ~60.30% accuracy—the exact baseline ratio.

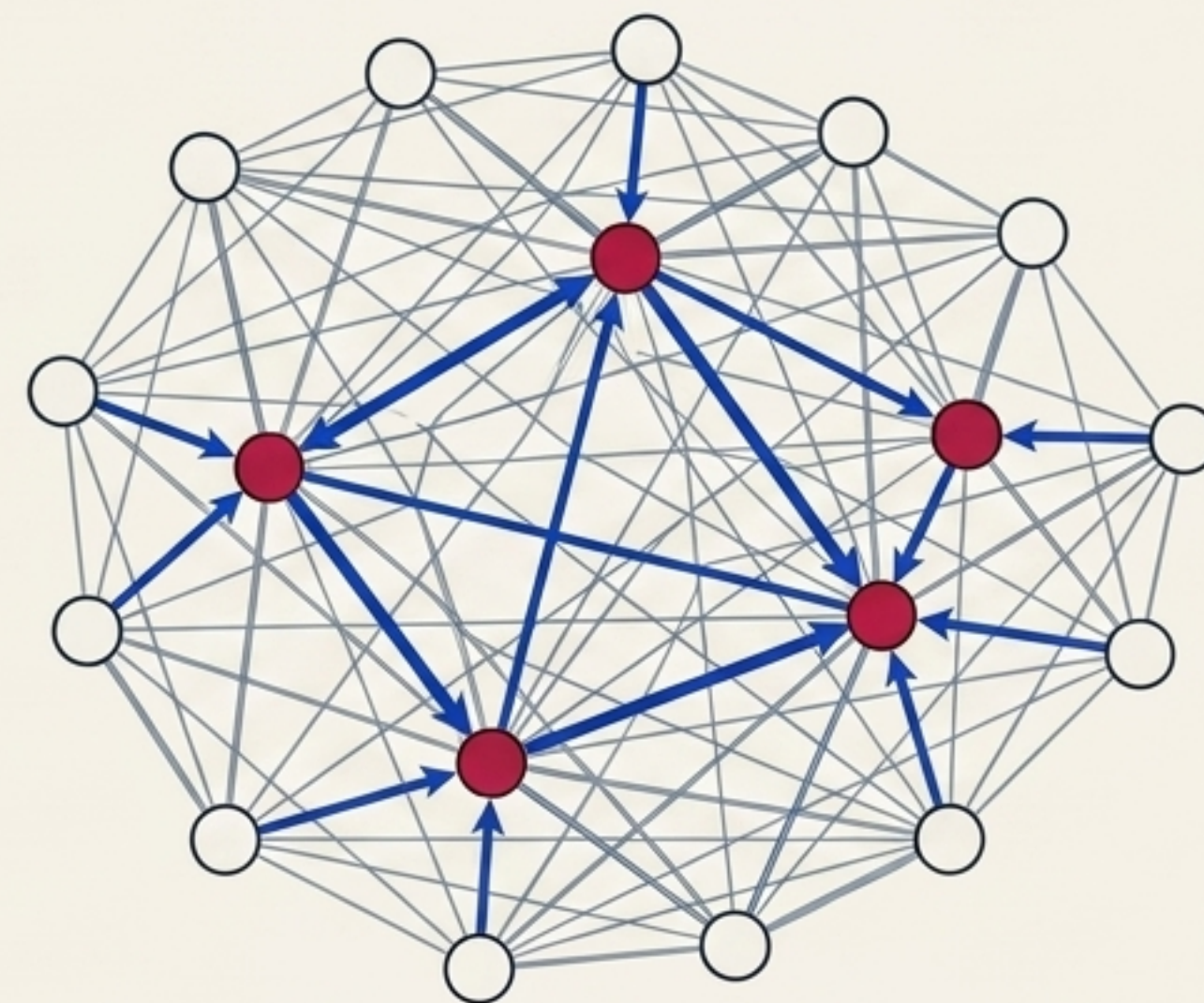
Searching for Non-Linear Synergies in Romance

PyTorch Attentive Tabular Network

Soft-Mask Routing: The network dynamically shifts priority across profile traits differently for every single user query.



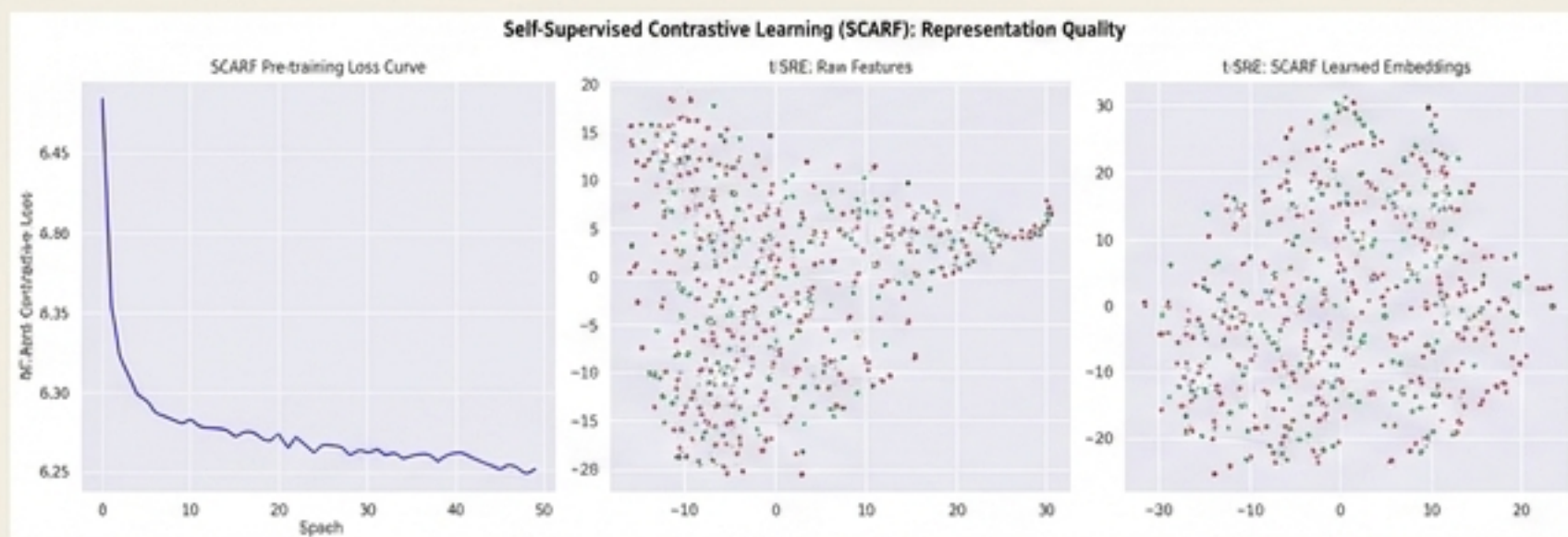
Graph Attention Networks (GAT)



Reframing the tabular problem as **semi-supervised node classification**, mapping **non-linear user-to-user** similarity graphs.

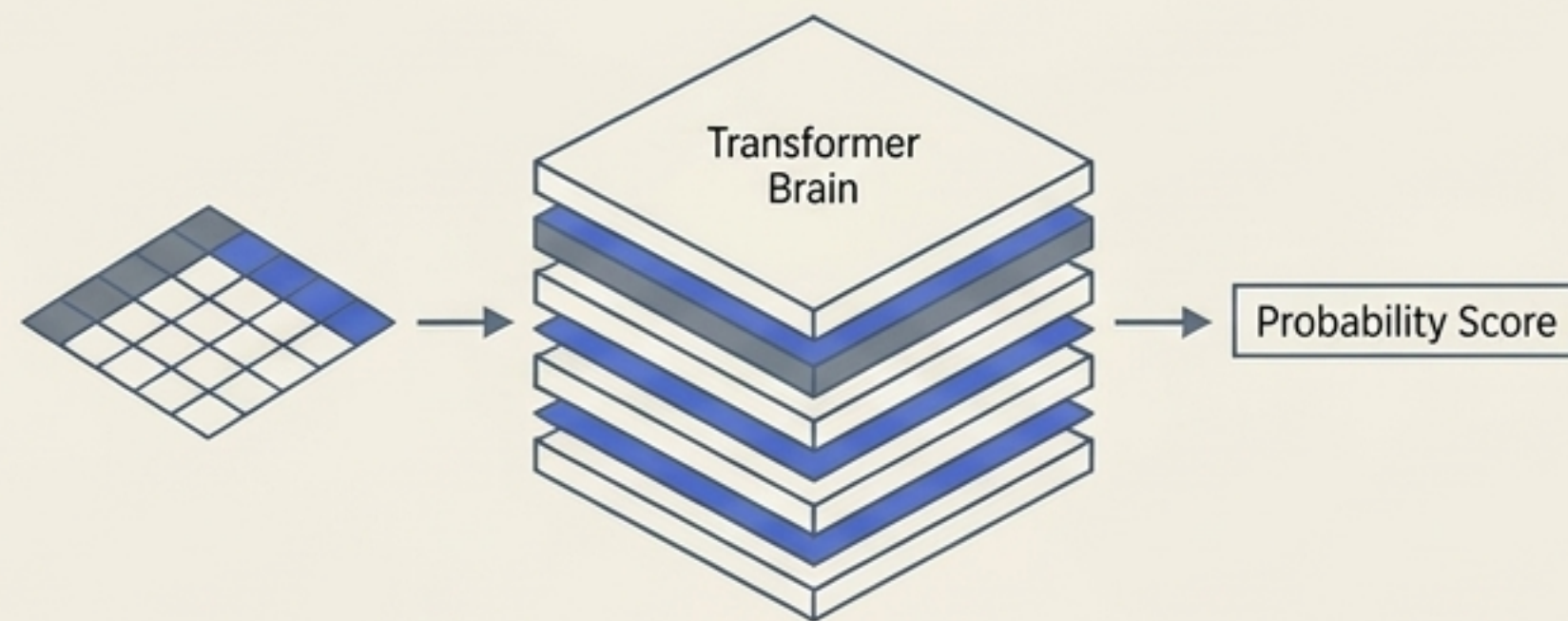
Transductive Pre-Training and Massive Priors

SCARF Contrastive Learning



t-SNE Representation: Latent space visualization reveals homogenous, overlapping clusters regardless of successful or failed match outcomes. Self-supervised pre-training via random feature corruption fails to force separation.

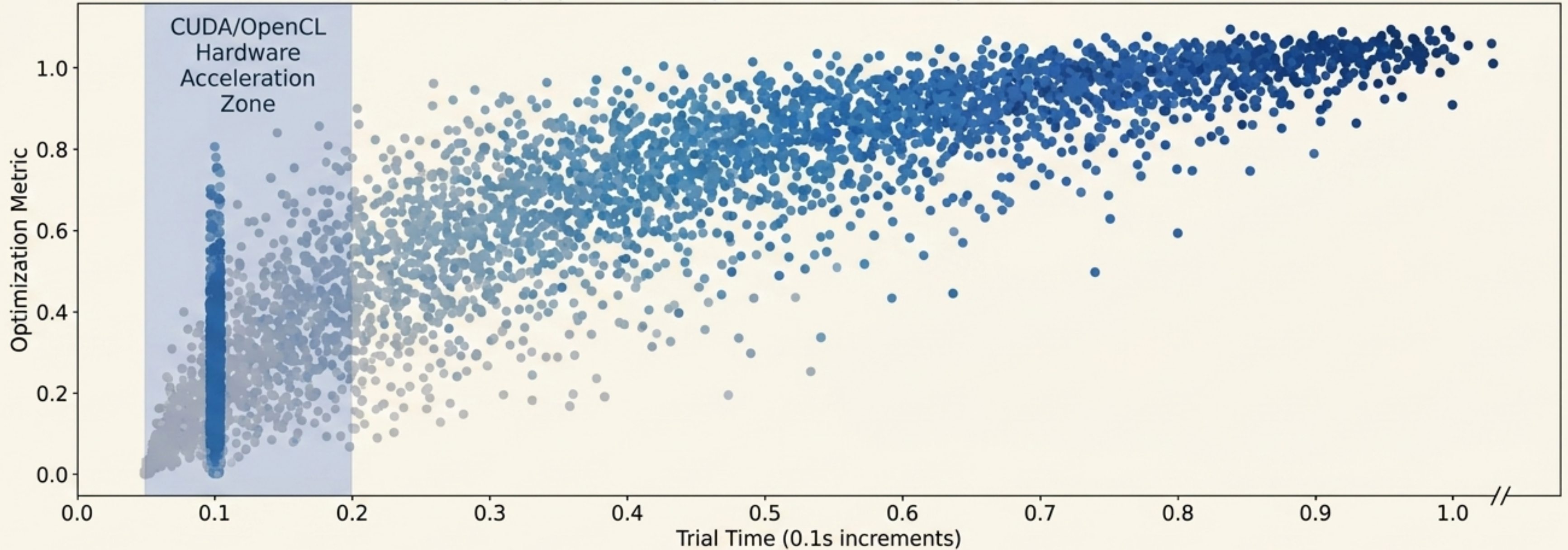
TabPFN Zero-Shot Evaluation



TabPFN Zero-Shot Evaluation: Bypassing gradient descent entirely using a Bayesian transformer pre-trained on millio: Bayesian transformer pre-trained on millions of synthetic datasets. It is the ultimate oracle, yet arrives at the exact same conclusion.

Accelerating 1,000 Optimization Trials to Under Four Minutes

Hyperparameter Optimization Landscape: 1,000 Trials



The Grid Search Bottleneck

Traditional hyperparameter tuning is catastrophically slow when exploring deep tree constraints.

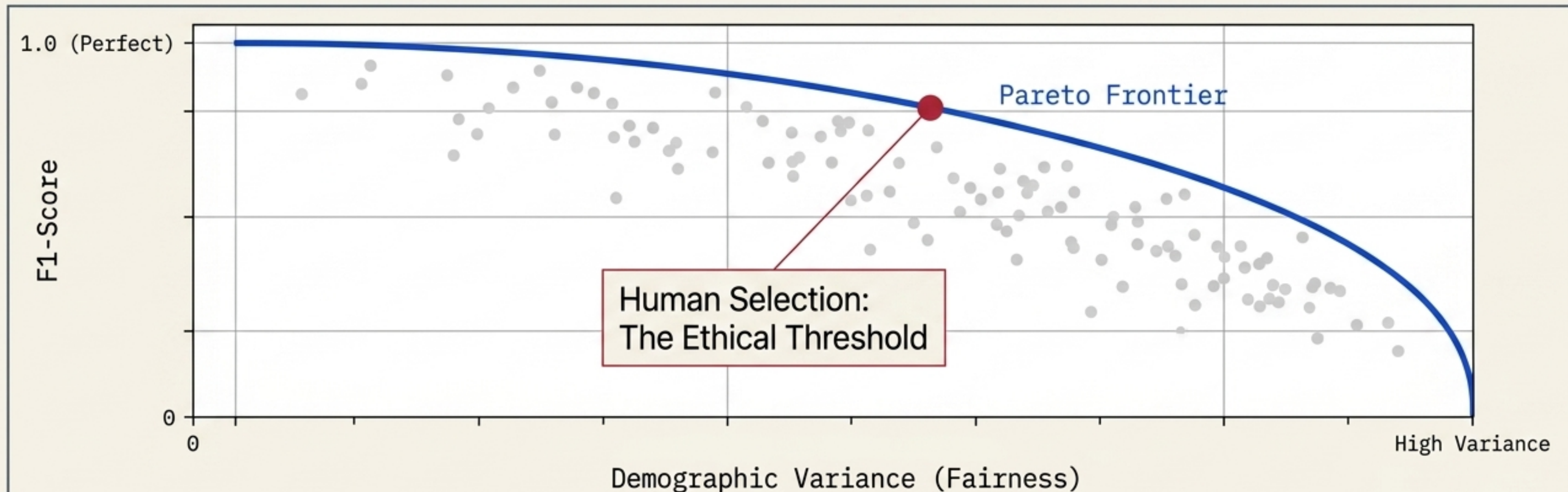
Massive Optuna Sweep

Executing a **1,000**-trial, fully randomized multi-objective optimization matrix.

CUDA/OpenCL Offloading

Routing trial-fitting calculations directly to graphics processing cores compresses individual trial latency down to **0.1 seconds**.

The Ethical Calculus of Predictive Power Versus Fairness



Conflicting Objectives

Maximizing predictive power (F1) inherently risks escalating algorithmic demographic bias.

The Pareto Frontier

Mapping the mathematical boundary where neither metric can improve without inherently degrading the other.

Ethical AI Engineering

Explicitly selecting an operating threshold that guarantees strict demographic parity across all user sub-groups.

Empirical Proof of the Absence of Predictive Signal

ROC-AUC ≈ 0.50

60.30%
Baseline



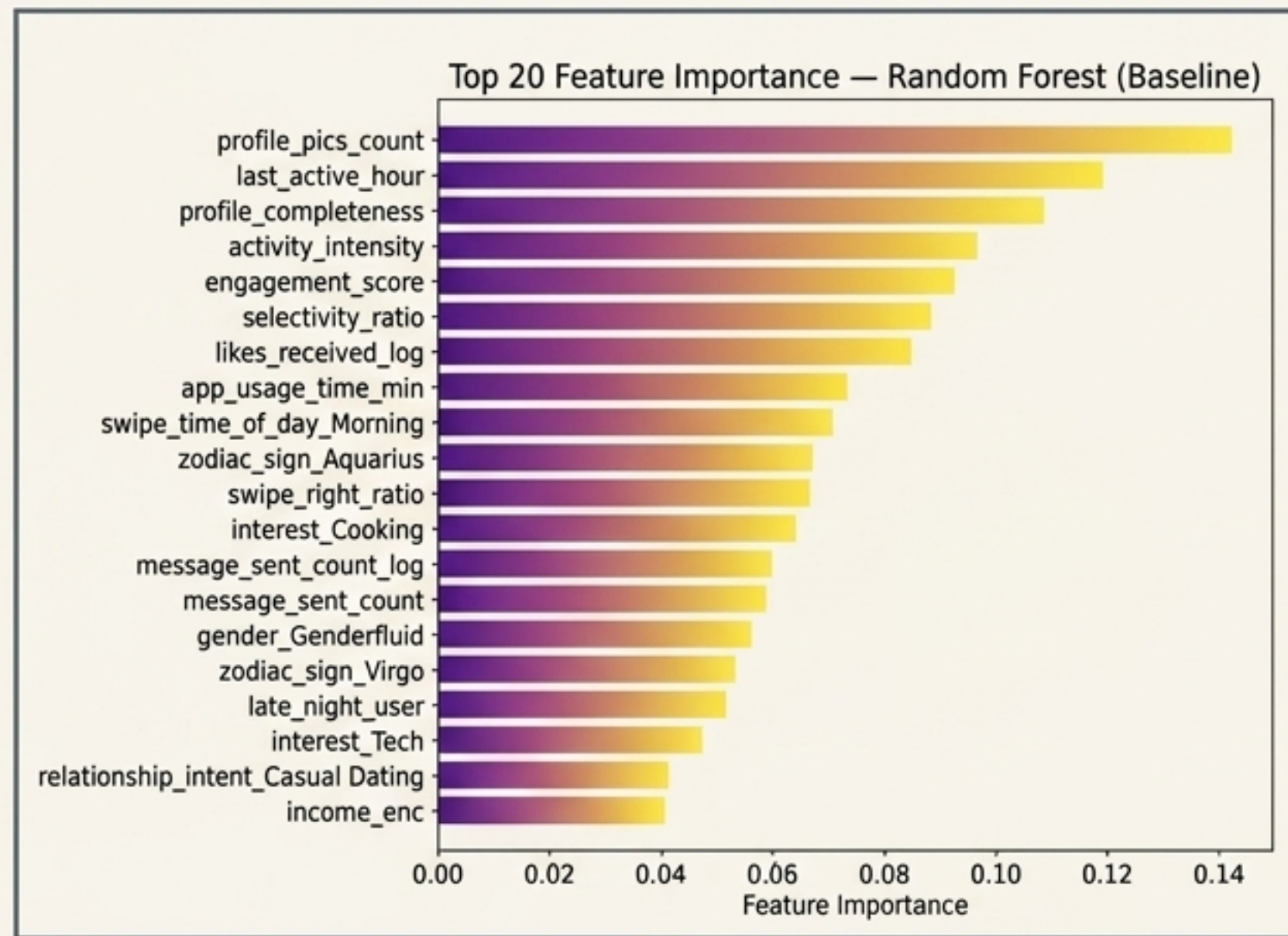
The Ultimate Convergence: All 16 models—from linear baselines to zero-shot tabular transformers—converge identically at ~60.30% test accuracy. The mathematical equivalent of a coin flip.

Scientific Honesty over Overfitting: There is absolutely no predictive signal in superficial profile features. Love cannot be predicted by swipe ratios, body types, or zodiac signs.

Prioritizing Explainability and Algorithmic Recourse



Rejecting the Black Box: With deep neural networks offering zero performance uplift, complex opacity becomes an unjustifiable architectural liability.

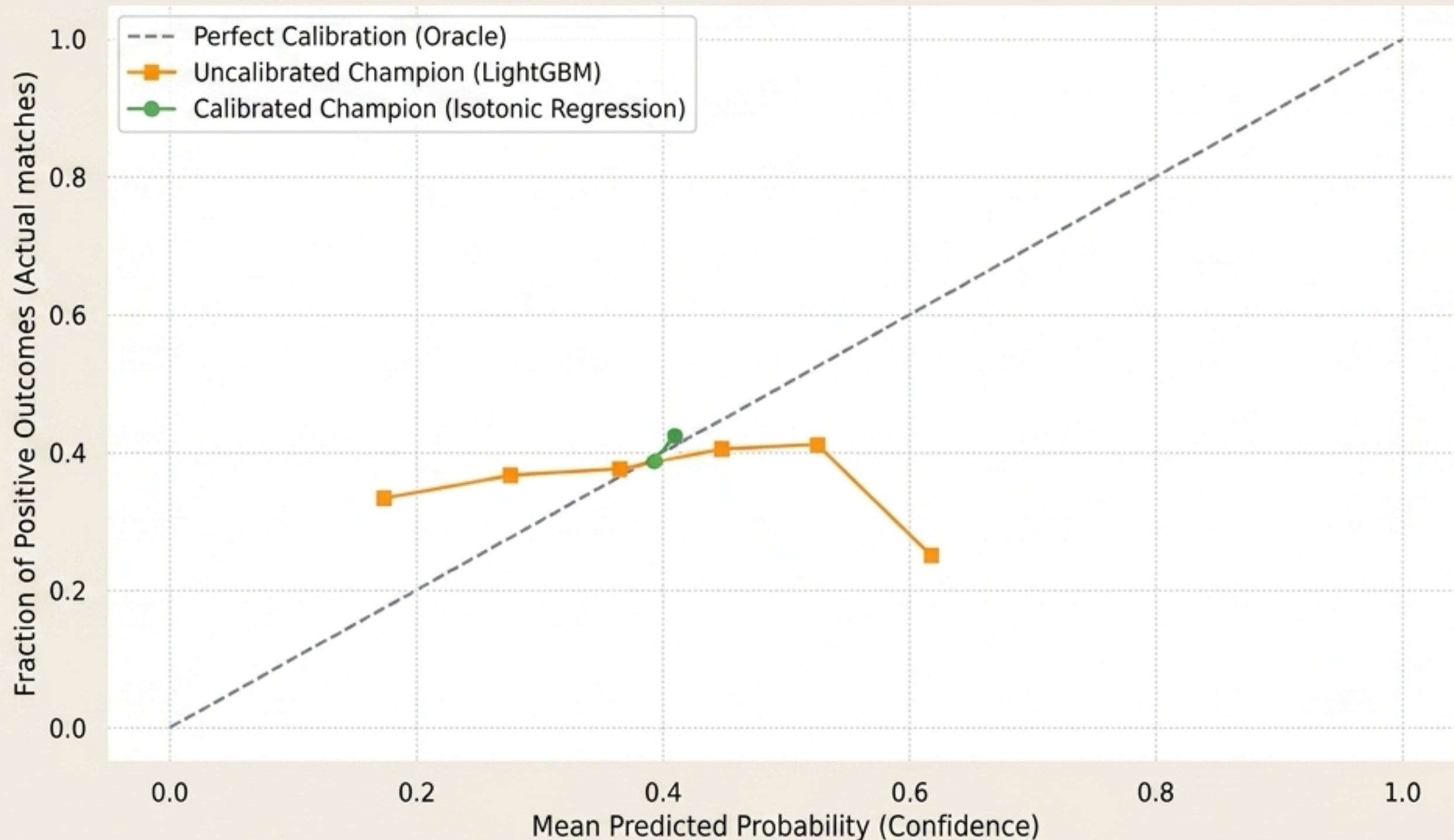


The Tree-Based Champion: Random Forest selected for its native architectural transparency.

Global Attribution & Recourse: Direct compatibility with SHAP values and Microsoft DiCE, providing users with actionable, understandable feedback on their profiles.

Aligning Raw Model Output with Empirical Reality

□ **Probability Calibration Curve (Reliability Diagram)**



The Calibration Gap

Raw output scores from tree-based algorithms are distorted margins, misrepresenting actual real-world probabilities.

Isotonic Regression

A strict, non-parametric mathematical mapping function applied globally over the Champion Model outputs.

Minimizing the Brier Score

Forcing predicted algorithmic confidence to perfectly mirror the empirical rate of connection success, guaranteeing trustworthy UI thresholds.